

Exam Date & Time: 29-May-2023 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

**FOURTH SEMESTER B.TECH. (ELECTRONICS AND COMMUNICATION ENGINEERING) DEGREE
EXAMINATIONS - MAY/JUNE 2023
SUBJECT: MAT 2257/MAT-2257 - ENGINEERING MATHEMATICS - IV
ENGINEERING MATHEMATICS IV [MAT 2257]**

Marks: 50**Duration: 180 mins.****A****Answer all the questions.**

- 1A) Solve the boundary value problem by finite difference method
 $y'' + x^2y = 0$, $y'(0) = 0$, $y'(1) = 1$ with $h = 0.5$. (4)
- 1B) Solve $u_{xx} + u_{yy} = -1$ in $|x| < 1$ and $|y| < 1$, given $u(x, \pm 1) = x^4$, $u(\pm 1, y) = y^4$ and $h = 0.5$. (3)
- 1C) Solve by Crank-Nicolson's method:
 $16u_t = u_{xx}$, $0 < x < 2$, $t > 0$. $u(x, 0) = 100\sin(\frac{\pi x}{2})$, $u(0, t) = u(2, t) = 100t$. (3)
 Compute u for two time steps with $h = 1/2$.
- 2A) An insurance company has insured 4000 doctors, 8000 teachers and 12000 businessmen. The chances of a doctor, teacher and businessmen dying before the age of 58 is 0.01, 0.03 and 0.05, respectively. If one of the insured people dies before 58, then find the probability that he is a doctor? (4)
- 2B) A man and a woman appear in an interview for two vacancies for the same post. The probability of man's selection is $1/3$ and the probability for the woman's selection is $1/2$. What is the probability that neither of them will be selected? (3)
- 2C) Find the straight line that best fits to the following data, using method of least squares:
- | | | | | | | | |
|--------|----|----|----|----|----|----|----|
| x | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| $f(x)$ | 14 | 27 | 40 | 55 | 68 | 77 | 85 |
- (3)
- 3A) The joint probability density function of X and Y is

$$f(x, y) = \begin{cases} 8xy, & 0 < x < y, 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$
 (4)
 (a) Find the $E(Y)$. (b) Determine $P(X + Y \leq 1)$. (c) Find Marginal Pdf of X .
- 3B) A fair coin is tossed thrice. Let $X = 0$ or 1 according as head or tail occurs on the first toss and $Y =$ number of heads. Compute the correlation coefficient between X and Y . (3)
- 3C) Let X be a one dimensional random variable with pdf

$$f(x) = \begin{cases} 4x - 4x^3, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$
 Find variance and mean of the random variable X . (3)
- 4A) The amount of time required to repair a TV is exponentially distributed with mean 2 hours. Find the following probabilities? (4)

- i) The probability that the required time exceeds 2 hours.
 ii) The conditional probability that the required time taken is at least 10 hours given that already 9 hours have been spent on repairing the TV.

4B) Let X be a random variable with the following pdf:

$$f(x) = \begin{cases} e^{-x}, & x > 0 \\ 0, & \text{Otherwise} \end{cases} \quad (3)$$

Find the moment generating function of X . Hence find mean $E(X)$ and Variance $V(X)$.

4C) Let X be a continuous random variable with the following pdf:

$$f(x) = \begin{cases} \frac{x}{12}, & 1 < x < 5 \\ 0, & \text{Otherwise} \end{cases} \quad (3)$$

Find the pdf of $Y = 2X - 3$?

5A) Solve the Difference equation $y_{n+2} - 2y_{n+1} + y_n = n^2 2^n$. (4)

5B) Solve the simultaneous equation using Z-transform
 $x_{n+1} - y_n = 1, y_{n+1} + x_n = 0, y_0 = -1, x_0 = 0$. (3)

5C) If X and Y are two independent random variable following normal distribution
 (i.e $X \sim N(0, \sigma^2), Y \sim N(0, \sigma^2)$) then find probability density function of $R = \sqrt{X^2 + Y^2}$. (3)

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